Security Log Detection System

Technical Architecture & Project Report

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Role: Cybersecurity R&D Intern | **Core Stack:** Python, Pandas, Scikit-Learn | **GitHub Repository:** github.com/ayah51/security-log-detection-system

1. EXECUTIVE SUMMARY & OBJECTIVES

Modern Security Operations Centers (SOCs) face significant alert fatigue due to high volumes of unaggregated authentication logs. Manually pinpointing cyber threats such as brute-force password guessing and distributed password spraying within raw security event feeds is inefficient and error-prone.

This project delivers a hybrid **Security Log Detection System** as a lightweight Minimum Viable Product (MVP). Implemented in Python, the system ingests structured CSV security logs, validates schema and address formatting integrity, executes deterministic rule-based threat detection using dynamic sliding windows, and applies an unsupervised Isolation Forest model to flag statistical behavioral anomalies at the IP level.

2. SYSTEM ARCHITECTURE & PROCESSING PIPELINE

The architecture follows a modular 4-stage pipeline ensuring independent validation, rapid rule processing, and low-latency alert generation:



- **Data Validation Engine:** Checks required fields (`timestamp` , `source_ip` , `username` , `event_type` , `status`), normalizes whitespace, parses ISO timestamps, verifies IPv4/IPv6 address syntax, and discards malformed records.
- **Rule-Based Detection Engine:** Applies time-bound window evaluations and account aggregation metrics to identify high-confidence signature patterns.
- **Machine Learning Anomaly Engine:** Evaluates behavioral metric distributions per source IP to surface complex threats.
- **Alert Management:** Formats detected threats with severity classifications, timestamps, and contextual rationale into `detection_alerts.csv` .

3. RULE-BASED DETECTION ENGINE MECHANICS

Rule 01: Repeated Failed Logins (Brute-Force Detection)

Monitors authentication failure events for each `source_ip` using a rolling 5-minute sliding time window. If the number of failed login attempts satisfies **f ≥ 5** within any 5-minute interval, the system generates a *Medium Severity* security alert.

Rule 02: Multiple Account Targeting (Password Spraying)

Tracks unique target usernames associated with failed authentication events per source IP. If a single IP targets **≥ 3 distinct accounts**, the activity is flagged to prevent single-account lockout evasion.

4. DETECTION ACCURACY & RELIABILITY MECHANISMS

To explicitly address detection reliability and minimize false positive / false negative rates, the engine integrates four core validation and precision safeguards:

- **Data Cleansing & Strict Validation:** Pre-filters malformed IP structures and corrupt timestamps, ensuring zero

5. UNSUPERVISED MACHINE LEARNING (ISOLATION FOREST)

To complement static rules, an **Isolation Forest** algorithm models multi-dimensional behavioral features per IP without requiring prior attack labels:

- **Engineered Features:** Total event volume, failed login count, successful login count, unique target usernames, event type diversity, and failure ratio (`failed_logins / total_events`).
- **Safety Safeguard:** To prevent statistical skew and false positives on small sample sizes, the ML engine automatically skips execution if the dataset contains fewer than 5 distinct IP addresses (`MIN_IPS_FOR_ML = 5`).

6. VERIFICATION, TESTING & EMPIRICAL ACCURACY RESULTS

The detection engine was verified against controlled synthetic test suites reproducing standard login behavior and deliberate attack vectors. All 5 core functional test cases achieved a **100% Precision & PASS rate**:

ID	Test Scenario	Input Conditions	Expected Output	Status
TC-01	Threshold Reached	5 failed logins from 1 IP within 5 mins	Alert generated (Medium Severity)	PASS
TC-02	Below Threshold	4 failed logins from 1 IP within 5 mins	No alerts generated	PASS
TC-03	Distributed IPs	5 failed logins across 3 distinct IPs	No single IP triggers threshold	PASS
TC-04	Outside Time Window	5 failed logins spread across 8 mins	Sliding window excludes old attempts	PASS
TC-05	Boundary Condition	Exactly 5 failed logins within 5:00 min	Alert generated precisely at threshold	PASS

7. SUMMARY OF KEY ACHIEVEMENTS

- **Reliable Threat Precision:** Zero false alarms during distributed or low-volume normal user login activity.
- **Resilient Input Handling:** Malformed lines and corrupt timestamps are isolated and logged without crashing the system.
- **Operational Accuracy:** Standardized CSV alert outputs allow immediate integration into SOC triage workflows with verified confidence.

8. FUTURE ROADMAP

1. **Real-Time Log Ingestion:** Upgrade static file parsing to stream processing using Apache Kafka and Syslog connectors.
2. **Automated SOAR Playbooks:** Integrate REST API endpoints to automatically trigger dynamic firewall IP blocking upon high-severity alerts.
3. **Threat Intelligence Enrichment:** Automatically cross-reference flagged IP addresses with external threat feeds (e.g., AbuseIPDB, VirusTotal).